$$I=\iint_D \frac{\cancel{2}x^2y+\cancel{2}y(4-x^2)}{\cancel{2}\sqrt{4-x^2}}\,dxdy$$

$$I=\iint_D \frac{\cancel{x^2y}+4y-\cancel{x^2y}}{\sqrt{4-x^2}}\,dxdy$$

$$I=\iint_D \frac{4y}{\sqrt{4-x^2}}\,dxdy = 16\pi$$

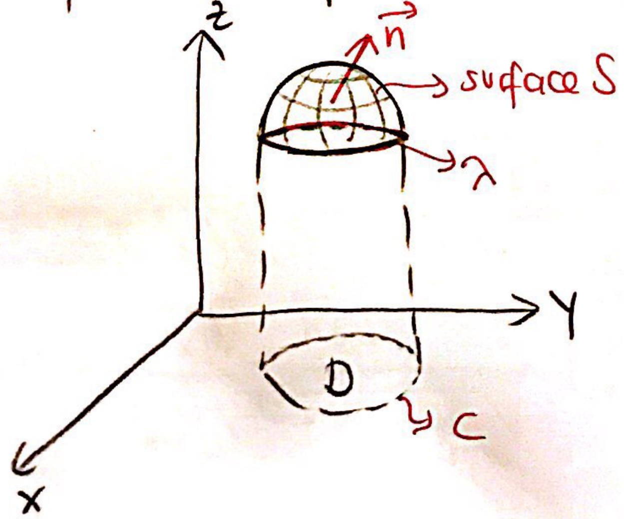
Stoke's Theorem

In this theorem, a line integral in space is transformed to a surface integral. Let the surface S is an open surface which is enclosed by the curve λ. For any point on the surface S, if the angle between normal line and coordinate axes are α, β, γ then;

$$\vec{n}=\cos\alpha\,\vec{i}+\cos\beta\,\vec{j}+\cos\gamma\,\vec{k}$$

$$\vec{n}=\frac{-\frac{\partial z}{\partial x}\vec{i}-\frac{\partial z}{\partial y}\vec{j}+\vec{k}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+1}}$$

Let $\vec{n}$ be the acute angle of positive normal of the surface and z-axis.



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Direction cosine of normal line of $z=f(x,y)$;

$$\boxed{\cos(\vec{n},x)=\cos\alpha=\frac{-\frac{\partial z}{\partial x}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+1}}}$$

$$\boxed{\cos(\vec{n},y)=\cos\beta=\frac{-\frac{\partial z}{\partial y}}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+1}}}$$

$$\boxed{\cos(\vec{n},z)=\cos\gamma=\frac{1}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+1}}}$$

Let the surface S is located on a space region V. Then consider the functions $L(x,y,z)$, $M(x,y,z)$ and $N(x,y,z)$ which are defined, continuous and have partial derivatives on this region. Let the projection of space curve λ be the curve C.

$$I=\int_{\lambda} L(x,y,z)\,dx \qquad \left(F(x,y,z)=0 \text{ or } z=f(x,y)\right)$$

$$\Rightarrow I=\int_{\lambda} L(x,y,f(x,y))\,dx$$

$$\oint_C P\,dx+Q\,dy=\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dx\,dy$$ by the Green's Formula;

then

$$\frac{\partial Q}{\partial x}=0 \Rightarrow I=\iint_D -\frac{\partial L}{\partial y}\,dx\,dy \; .$$

$$L=L(x,y,z) \rightarrow \left.\begin{array}{l} x=x \\ y=y \\ z=f(x,y) \end{array}\right\} \frac{\partial L}{\partial y}=\frac{\partial L}{\partial x}\cdot\overset{0}{\cancel{\frac{\partial x}{\partial y}}}+\frac{\partial L}{\partial y}\cdot\overbrace{\frac{\partial y}{\partial y}}^{1}+\frac{\partial L}{\partial z}\cdot\frac{\partial z}{\partial y}$$

$$\Rightarrow I=\iint_D -\left(\frac{\partial L}{\partial y}-\frac{\partial L}{\partial z}\cdot\frac{\partial z}{\partial y}\right)dx\,dy$$

$$I=\iint_D -\frac{\partial L}{\partial y}\,dx\,dy+\iint_D \frac{\partial L}{\partial z}\cdot\frac{\partial z}{\partial y}\,dx\,dy$$

$$ds=\sec\gamma\,dx\,dy \quad \Rightarrow dx\,dy=\cos\gamma\,ds$$

$$I=\iint_D -\frac{\partial L}{\partial y}\cos\gamma\,ds+\iint_D \frac{\partial L}{\partial z}\cdot\frac{\partial z}{\partial y}\cdot\cos\gamma\,ds$$

if $\dfrac{\cos\beta}{\cos\gamma}=\dfrac{\partial z}{\partial y}$;

$$I=\iint_D -\frac{\partial L}{\partial y}\cos\gamma\,ds+\iint_D \frac{\partial L}{\partial z}\cdot\cos\beta\,ds$$

$$\boxed{I=\int_\lambda L(x,y,z)\,dx=-\iint_D \frac{\partial L}{\partial y}\cos\gamma\,ds+\iint_D \frac{\partial L}{\partial z}\cos\beta\,ds}$$

1

(*) $I = \int_{\lambda} M(x,y,z)dy = \int_{\lambda} M(x,y,f(x,y))dy = \iint_D \frac{\partial M}{\partial x} dxdy$

$$\left.\begin{cases} x = x \\ y = y \\ z = f(x,y) \end{cases} \quad \int_{\lambda} M(x,y,f(x,y))dy \right\}$$

Since $\frac{\partial M}{\partial x} = \frac{\partial M}{\partial x} + \frac{\partial M}{\partial z} \cdot \frac{\partial z}{\partial x}$, then

$$I = \iint_D \left(\frac{\partial M}{\partial x} + \frac{\partial M}{\partial z} \cdot \frac{\partial z}{\partial x} \right) dxdy = \iint_D \frac{\partial M}{\partial x} dxdy + \iint_D \frac{\partial M}{\partial z} \cdot \frac{\partial z}{\partial x} dxdy$$

$$\left.\begin{aligned} ds &= \sec\gamma\, dxdy \\ dxdy &= \cos\gamma\, ds \end{aligned}\right\} \; I = \iint_S \frac{\partial M}{\partial x} \cos\gamma\, ds + \iint_S \frac{\partial M}{\partial z} \cdot \frac{\partial z}{\partial x} \cos\gamma\, ds$$

Since $-\frac{\partial z}{\partial x} = \frac{\cos\alpha}{\cos\gamma}$, (*) then

$I = \iint_S \frac{\partial M}{\partial x} \cos\gamma\, ds - \iint_S \frac{\partial M}{\partial z} \cos\alpha\, ds$. So

$$\boxed{I = \int_{\lambda} M(x,y,z)dy = \iint_S \frac{\partial M}{\partial x} \cos\gamma\, ds - \iint_S \frac{\partial M}{\partial z} \cos\alpha\, ds} \quad (2)$$

(*) Now projection will be on xz-plane. So the function will be $y = f(x,z)$.

$$I=\int_{\lambda} N(x,y,z)\,dz=\int_{\lambda} N(x,f(x,z),z)\,dz=\iint_{D} -\frac{\partial N}{\partial x}\,dx\,dz$$

$$\frac{\partial N}{\partial x}=\frac{\partial N}{\partial x}+\frac{\partial N}{\partial y}\cdot\frac{\partial y}{\partial x}$$ then;

$$I=\iint_{D} -\left(\frac{\partial N}{\partial x}+\frac{\partial N}{\partial y}\cdot\frac{\partial y}{\partial x}\right)dx\,dz=\iint_{D} -\frac{\partial N}{\partial x}\,dx\,dz+\iint_{N} -\frac{\partial N}{\partial y}\cdot\frac{\partial y}{\partial x}\,dx\,dz$$

$$\left\{\oint_{C} Q\,dz+P\,dx=\iint_{D}\left(\frac{\partial P}{\partial z}-\frac{\partial Q}{\partial x}\right)dx\,dz\right\}$$

$$\left.\begin{array}{l} ds=\sec\beta\,dx\,dz \\ dx\,dz=\cos\beta\,ds \end{array}\right\} I=\iint_{S} -\frac{\partial N}{\partial x}\cos\beta\,ds+\iint_{S} -\frac{\partial N}{\partial y}\cdot\frac{\partial y}{\partial x}\cos\beta\,ds$$

$$\vec{n}=\cos\alpha\,\vec{i}+\cos\beta\,\vec{j}+\cos\gamma\,\vec{k}$$

$$y=f(x,z)\Rightarrow F=y-f(x,z)=0$$

$$\vec{n}=\frac{-\frac{\partial f}{\partial x}\vec{i}+\vec{j}-\frac{\partial f}{\partial z}\vec{k}}{\sqrt{\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial z}\right)^2+1}}\Rightarrow\frac{\partial y}{\partial x}=-\frac{\cos\alpha}{\cos\beta}$$

$$\boxed{I=\int_{\lambda} N(x,y,z)\,dz=-\iint_{S}\frac{\partial N}{\partial x}\cos\beta\,ds+\iint_{S}\frac{\partial N}{\partial y}\cos\alpha\,ds}\quad (3)$$

① + ② + ③ = I

$$I = \int_{\lambda} L(x,y,z)dx + M(x,y,z)dy + N(x,y,z)dz$$

$$I = \iint_S \left(\left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial z}\right)\cos\alpha + \left(\frac{\partial L}{\partial z} - \frac{\partial N}{\partial x}\right)\cos\beta + \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y}\right)\cos\gamma \right) ds$$

Stoke's Formula

⊛ $\left.\begin{aligned} \vec{F} &= L\vec{i} + M\vec{j} + N\vec{k} \\ d\vec{r} &= dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k} \end{aligned}\right\}$ then

$$\boxed{\int_{\lambda} \vec{F} d\vec{r} = \iint_S \text{rot}\vec{F}.\vec{n}\,ds}$$

Vector form of Stoke's Formula

$$\text{rot}\,\vec{F} = \nabla \wedge \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ L & M & N \end{vmatrix}$$

$$\text{rot}\vec{F} = \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial z}\right)\vec{i} + \left(\frac{\partial L}{\partial z} - \frac{\partial N}{\partial x}\right)\vec{j} + \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y}\right)\vec{k}$$

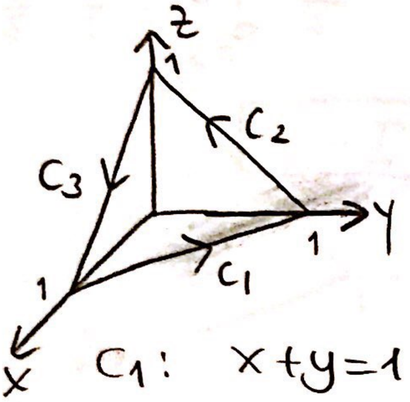
$$\vec{n} = \cos\alpha\,\vec{i} + \cos\beta\,\vec{j} + \cos\gamma\,\vec{k}$$

Note: If the surface S is a parallel plane to xy-plane then Stoke's Formula becomes Green's Formula.

<u>Example</u>: $\vec{F} = (2y+z)\vec{i} + (x-z)\vec{j} + (y-4)\vec{k}$ is given. Let

c be enclosing the triangle which is generated by the intersection of the plane $x+y+z=1$ and coordinate planes. Verify Stoke's formula.

$$I = \int_C \vec{F}d\vec{r} = \iint_S \text{Rot}\vec{F} \cdot \vec{N}ds$$



$C: C_1 + C_2 + C_3$

$$I = \int_C \vec{F}d\vec{r} = \int_C (2y+z)dx + (x-z)dy + (y-4)dz$$

C_1: $x+y=1 \Rightarrow \begin{matrix} x=x \\ y=1-x \\ z=0 \end{matrix} \left\{ \begin{matrix} dx=dx \\ dy=-dx \\ dz=0 \end{matrix} \right.$

$$I_1 = \int_1^0 (2-2x+0)dx + (x-0)(-dx) = \int_0^1 (2-2x-x)dx$$

$$I_1 = \int_1^0 (2-3x)dx = 2x - \frac{3}{2}x^2 \Big|_1^0 = \frac{3}{2} - 2 = -\frac{1}{2} \;//$$

C_2: $y+z=1 \Rightarrow \left. \begin{matrix} x=0 \\ y=y \\ z=1-y \end{matrix} \right\} \begin{matrix} dx=0 \\ dy=dy \\ dz=-dy \end{matrix}$

$$I_2 = \int_1^0 (2y+1-y) \cdot 0 + (0-1+y)dy + (y-4)(-dy)$$

$$I_2 = \int_1^0 (-1+y-y+4)dy = 3\int_1^0 dy = -3 \;//$$

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$$C_3: x+z=1 \Rightarrow \left.\begin{array}{l} x=x \\ y=0 \\ z=1-x \end{array}\right\} \begin{array}{l} dx=dx \\ dy=0 \\ dz=-dx \end{array}$$

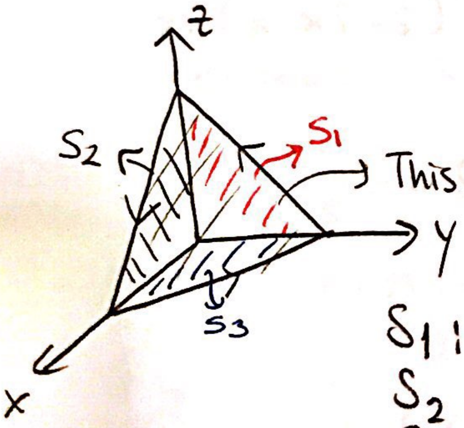
$$I_3 = \int_0^1 (1-x)dx + (x-1+x).0 + (0-4)(-dx)$$

$$I_3 = \int_0^1 (1-x+4)dx = \int_0^1 (5-x)dx = 5x - \frac{x^2}{2}\Big|_0^1 = 5 - \frac{1}{2} = \frac{9}{2} \,//$$

$$I = I_1 + I_2 + I_3 = -\frac{1}{2} - 3 + \frac{9}{2} = 1 \Rightarrow \boxed{I=1}$$

⊛ $\vec{F} = (2y+z)\vec{i} + (x-z)\vec{j} + (y-4)\vec{k}$

$$\text{rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2y+z & x-z & y-4 \end{vmatrix} = (1+1)\vec{i} + (1-0)\vec{j} + (1-2)\vec{k} = 2\vec{i} + \vec{j} - \vec{k}$$



→ $(x+y+z=1)$

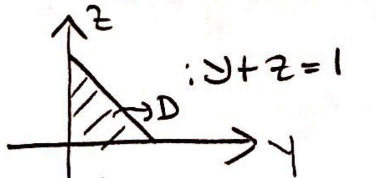
→ This surface is open! So, 3 surfaces we have to evaluate surface intg.

$S_1: x=0$ (yz-plane)
$S_2: y=0$ (xz-plane)
$S_3: z=0$ (xy-plane)

S_1: $x=0$ $\quad \vec{N} = -\vec{i}$, $\quad ds = \underbrace{\sec\alpha}_{1} dydz \Rightarrow ds = dydz$

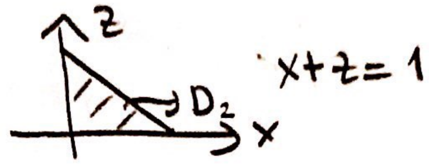
$$I_1 = \iint_{S_1} \text{Rot}\vec{F}.\vec{N} ds = \iint_D (2\vec{i} + \vec{j} - \vec{k})(-\vec{i}) dydz$$

$$I_1 = -2\iint_D dydz$$



$$I_1 = -2\int_0^1 \int_0^{1-y} dz dy = -2\int_0^1 (1-y)dy = -2\left(y - \frac{y^2}{2}\right) = -1 \,//$$

S_2 : $y=0$ $\quad \vec{N} = -\vec{j}$



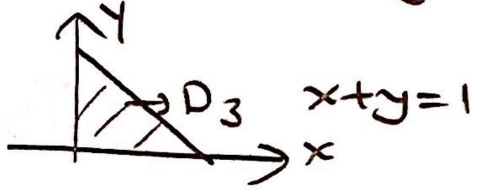
$ds = \sec\beta \, dx dz = dx dz$

$$I_2 = \iint_{S_2} Rot\vec{F} \cdot \vec{N} ds = \iint_{D_2} (2\vec{i} + \vec{j} - \vec{k})(-\vec{j}) dx dz$$

$$I_2 = -\iint_{D_2} dx dz = -\frac{1}{2} \,//$$

S_3 : $z=0$ $\quad \vec{N} = -\vec{k}$ $\quad ds = \sec\gamma \, dx dy = dx dy$

$$I_3 = \iint_{S_3} Rot\vec{F} \cdot \vec{N} ds = \iint_{D_3} (2\vec{i} + \vec{j} - \vec{k})(-\vec{k}) dx dy$$



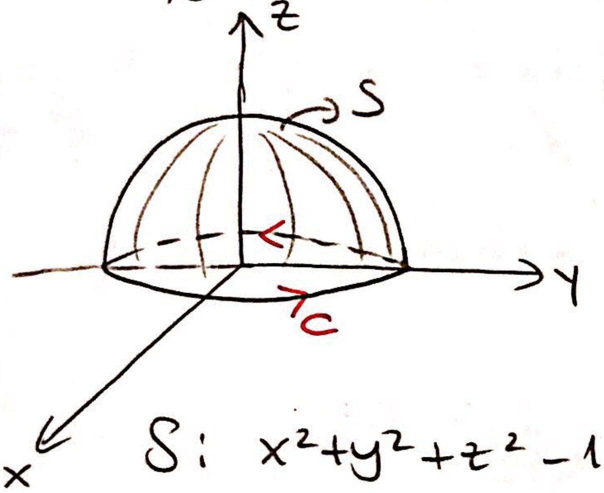
$$I_3 = \iint_{D_3} dx dy = \frac{1}{2} \,//$$

$$I = I_1 + I_2 + I_3 = -1 - \frac{1}{2} + \frac{1}{2} = \boxed{-1}$$

Example: $S: \left.\begin{array}{l} x^2+y^2+z^2=1 \\ z \geq 0 \end{array}\right\}$ and $\vec{F} = y^2\vec{i} + xy\vec{j} + xz\vec{k}$

are given. Then a) Evaluate $I = \iint_S \mathrm{Rot}\vec{F}\cdot\vec{n}\,ds$

b) Verify Stokes Formula.



$$\mathrm{rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & xy & xz \end{vmatrix}$$

$$\mathrm{rot}\vec{F} = 0.\vec{i} - z\vec{j} + (y-2y)\vec{k}$$

$$\mathrm{rot}\vec{F} = -z\vec{j} - y\vec{k}$$

$S: x^2+y^2+z^2-1=0$

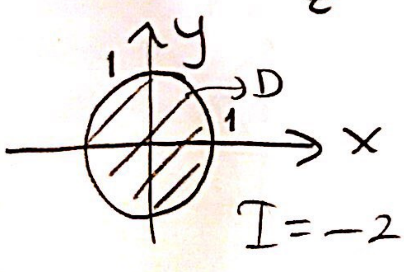
$$\vec{n} = \frac{2x\vec{i}+2y\vec{j}+2z\vec{k}}{\sqrt{4x^2+4y^2+4z^2}} = \frac{x\vec{i}+y\vec{j}+z\vec{k}}{\sqrt{x^2+y^2+z^2}}$$

$$ds = \sec\gamma\,dx\,dy \quad , \quad \vec{n}.\vec{k} = \frac{z}{\sqrt{x^2+y^2+z^2}} = \cos\gamma$$

$$\sec\gamma = \frac{\sqrt{x^2+y^2+z^2}}{z}$$

$$I = \iint_S \mathrm{Rot}\vec{F}.\vec{n}\,ds = \iint_S (-z\vec{j}-y\vec{k})\frac{(x\vec{i}+y\vec{j}+z\vec{k})}{\cancel{\sqrt{x^2+y^2+z^2}}}.\frac{\cancel{\sqrt{x^2+y^2+z^2}}}{z}.dxdy$$

$$I = \iint_D \frac{(-zy-yz)}{z}dxdy = \iint_D -2\frac{y\cancel{z}}{\cancel{z}}dxdy = -2\iint_D y\,dxdy$$



$$I = -2\int_0^{2\pi}\int_0^1 r\sin\theta\, r\,dr\,d\theta$$

$$I = -2\int_0^{2\pi}\int_0^1 \sin\theta . r^2 dr\,d\theta = -\frac{2}{3}\int_0^{2\pi} \sin\theta . r^3\Big|_0^1 d\theta$$

$$I = -\frac{2}{3}\int_0^{2\pi}\sin\theta\,d\theta = \frac{2}{3}\cos\theta\Big|_0^{2\pi} = \frac{2}{3}(1-1) = 0$$

b) Stoke's Formula; $\int_C \vec{F}.d\vec{r} = \iint_S \text{Rot}\vec{F}.\vec{n}\,ds$

$C: x^2+y^2=1$, $\left.\begin{array}{l} x=\cos t \\ y=\sin t \\ z=0 \end{array}\right\} \begin{array}{l} dx=-\sin t\,dt \\ dy=\cos t\,dt \\ dz=0 \end{array}$

$I=\int_C \vec{F}.d\vec{r} = \int_C y^2dx + xydy + xzdz$

$I = \int_0^{2\pi} (\sin^2 t)(-\sin t)dt + (\cos t.\sin t)\cos t\,dt + 0$

$I = \int_0^{2\pi} (-\sin^3 t + \cos^2 t.\sin t)dt = \underbrace{-\int_0^{2\pi} \sin^3 t\,dt}_{I_1} +$

$\int_0^{2\pi} \underbrace{\cos^2 t.\sin t\,dt}_{I_2}$

$I_1 = -\int_0^{2\pi} \sin^3 t\,dt = -\int_0^{2\pi} \sin t.\cos^2 t\,dt$, $\cos t = u$
$-\sin t\,dt = du$

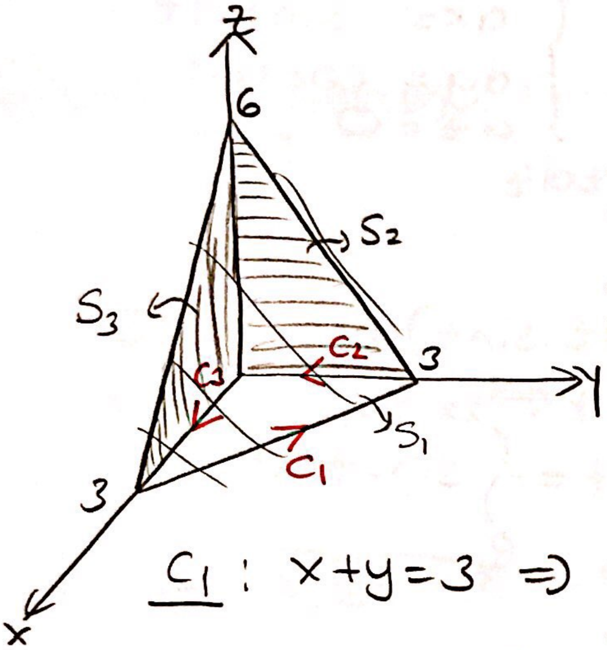
$I_1 = \int_1^1 u^2 du = 0$

$I_2 = \int_0^{2\pi} \cos^2 t.\sin t\,dt$, $\cos t = u$
$-\sin t\,dt = du$

$I_2 = 0$

$\int_C \vec{F}d\vec{r} = 0$ //

Example: $S:\ \left.\begin{array}{l} 2x+2y+z=6 \\ y=0,\ x=0 \end{array}\right\}$ and $\vec{F}=xy\vec{i}-y^2\vec{j}+(x+z)\vec{k}$ are given. Verify Stoke's Formula.



$C: C_1+C_2+C_3$

$$I=\int_C \vec{F}.d\vec{r}$$

$$I=\int_C xy\,dx-y^2dy+(x+z)dz$$

$\underline{C_1}:\ x+y=3 \Rightarrow \left.\begin{array}{l} x=x \\ y=3-x \\ z=0 \end{array}\right\} \begin{array}{l} dx=dx \\ dy=-dx \\ dz=0 \end{array}$

$$I_1=\int_3^0 x(3-x)dx+(3-x)^2dx+0$$

$$I_1=\int_3^0 (3x-\cancel{x^2}+9-6x+\cancel{x^2})\,dx=\int_3^0(-3x+9)\,dx \Rightarrow$$

$$I_1=-\frac{3}{2}x^2+9x\Big|_3^0=0-\left(-\frac{27}{2}+27\right)=-\frac{27}{2}\ //$$

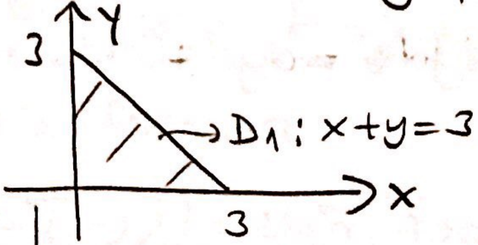
$\underline{C_2}:\ \left.\begin{array}{l} x=0 \\ y=y \\ z=0 \end{array}\right\} \begin{array}{l} dx=0 \\ dy=dy \\ dz=0 \end{array}$ $\quad I_2=\int_3^0 -y^2dy=-\frac{y^3}{3}\Big|_3^0=9\ //$

$\underline{C_3}:\ \left.\begin{array}{l} x=x \\ y=0 \\ z=0 \end{array}\right\} \begin{array}{l} dx=dx \\ dy=0 \\ dz=0 \end{array}$ $\quad I_3=\int_0^3 x.0.dx=0\ //$

$$I=I_1+I_2+I_3=9-\frac{27}{2}+0=-\frac{9}{2}\ //$$

$\underline{S_1}$: $2x+2y+z=6$

projection on xy-plane



$$\mathrm{Rot}\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & -y^2 & x+z \end{vmatrix} = 0.\vec{i} - \vec{j} - x\vec{k}$$

$$\mathrm{Rot}\vec{F} = -\vec{j} - x\vec{k}$$

$$\vec{n} = \frac{2\vec{i}+2\vec{j}+\vec{k}}{\sqrt{9}} = \frac{2\vec{i}+2\vec{j}+\vec{k}}{3}$$

$$\vec{n}.\vec{k} = \frac{1}{3} = \cos\gamma \Rightarrow \sec\gamma = 3$$

$$ds = 3dxdy$$

$$I_1 = \iint_{S_1} \mathrm{Rot}\vec{F}.\vec{n}\,ds = \iint_{D_1} (-\vec{j}-x\vec{k})\left(\frac{2\vec{i}+2\vec{j}+\vec{k}}{\cancel{3}}\right).\cancel{3}dxdy$$

$$I_1 = \iint_{D_1}(-2-x)dxdy = -\iint_{D_1}(2+x)dxdy$$

$$I_1 = -\int_0^3\int_0^{3-y}(2+x)dxdy = -\int_0^3 2x+\frac{x^2}{2}\Big|_0^{3-y}dy$$

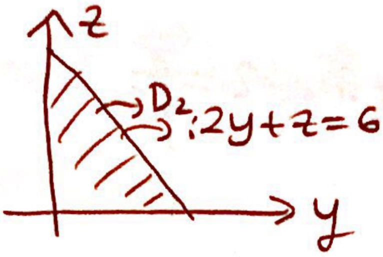
$$I_1 = -\int_0^3\left[2(3-y)+\frac{(3-y)^2}{2}\right]dy = -\int_0^3\left(6-2y+\frac{9}{2}-3y+\frac{y^2}{2}\right)dy$$

$$I_1 = \int_0^3\left(-\frac{y^2}{2}+5y-\frac{21}{2}\right)dy = -\frac{y^3}{6}+\frac{5}{2}y^2-\frac{21y}{2}\Big|_0^3$$

$$I_1 = -\frac{27}{6}+\frac{5}{2}.9-\frac{63}{2} = -\frac{27}{2} \,//$$

S_2: $x=0$, $\vec{n}=-\vec{i}$

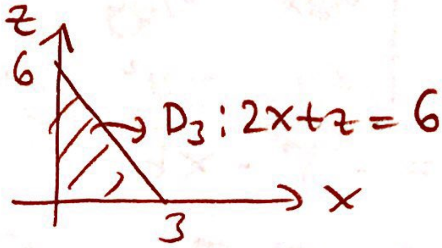
$ds=\underbrace{\sec\alpha}_{1}\, dydz = dydz$



$$I_2=\iint_{S_2} Rot\vec{F}.\vec{n}ds=\iint_{D_2}(-\vec{j}-x\vec{k})(-\vec{i}).\, dydz = 0$$

S_3: $y=0$, $\vec{n}=-\vec{j}$

$\vec{n}.\vec{j}=1=\sec\beta=\cos\beta$



$$I_3=\iint_{S_3} Rot\vec{F}.\vec{n}ds=\iint_{D_3}(-\vec{j}-x\vec{k}).(-\vec{j}).\, dxdz$$

$$I_3=\iint_{D_3} dzdx=\int_0^3\int_0^{6-2x} dzdx=\int_0^3 z\Big|_0^{6-2x}dx=\int_0^3(6-2x)dx$$

$$I_3=(6x-x^2)\Big|_0^3 = 18-9=9$$

for; $S_1+S_2+S_3 \Rightarrow I=I_1+I_2+I_3=-\frac{27}{2}+0+9=-\frac{9}{2}$